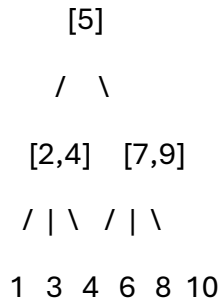


SQL Server Questions

53. Clustered vs Non-Clustered Index.

A **Clustered Index** stores the actual table data in sorted order.

The table itself is organized according to the indexed column.



SQL Server can quickly jump to the correct location using a B-Tree instead of scanning every row.

A Non-Clustered Index is a separate lookup structure that stores indexed values and pointers to the actual rows. For example, if I create a non-clustered index on the `Email` column, SQL Server stores email values in a B-Tree along with references to the corresponding rows. When I search by email, SQL Server first searches the index, finds the pointer, and then retrieves the actual row from the table. This avoids scanning the entire table and significantly improves query performance.

54. What is index fragmentation?

Index Fragmentation means the pages of an index become **out of order or contain empty space**, making SQL Server read more pages than necessary

Index fragmentation occurs when the logical order of index pages no longer matches their physical order or when pages contain excessive empty space due to inserts, updates, deletes, and page splits.

How to Fix Fragmentation

If fragmentation is between 5% and 30%

Use:

```
ALTER INDEX IX_Customers_Email
ON Customers
REORGANIZE;
```

This rearranges pages without recreating the index.

If fragmentation is greater than 30%

Use:

```
ALTER INDEX IX_Customers_Email  
ON Customers  
REBUILD;
```

This drops and recreates the index.

55. Explain execution plans.

56. Difference between:

DELETE-We can use where

DELETE FROM Customers;(Slower)

Identity is keeping

TRUNCATE- Identity Reset - we cannot use where

DROP- Removes the entire object.

57. What is a deadlock?

When Transaction A is waiting for the resource that is locked by Transaction B.

And Transaction B is waiting for the resource that is locked by transaction A

SQL Server detects this situation and automatically kills one transaction (called the **deadlock victim**) to break the cycle and allow other transaction to proceed.

58. How do you identify slow queries?

59. Difference between:

INNER JOIN

LEFT JOIN

RIGHT JOIN

FULL JOIN

60. What are CTEs?

61. What are window functions?

Example:

ROW_NUMBER()

RANK()

DENSE_RANK()

62. What is parameter sniffing?

This is a very common senior-level SQL interview question.

63. How do you optimize stored procedures?